Detecting anti-holomorphic dynamics via symbolic regression with the eml* operator

Anthony Monnerot
Independent researcher, France
`github.com/antparis/oxieml-star`

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Abstract

We introduce $\mathrm{eml}^\star(x, y) = \exp(\overline{x}) - \ln(\overline{y})$, the anti-holomorphic extension of the EML operator $\mathrm{eml}(x, y) = \exp(x) - \ln(y)$ of Odrzywołek (2026). We prove that $\overline{z} = 1 - \mathrm{eml}^\star(0, \mathrm{eml}(z, 1))$ (Theorem 1), providing a purely algebraic representation of complex conjugation within the EML framework. Combined with symbolic regression (PySR), this yields an automatic criterion for classifying complex dynamical systems: eml* appears in the discovered equation when the underlying dynamics is anti-holomorphic and does not appear when it is holomorphic, across all tested systems. We validate this on three domains: (i) fractal iteration maps, where PySR distinguishes the Mandelbrot set ($z^2 + c$, holomorphic, eml*-free) from the Tricorn ($\overline{z}^2 + c$, anti-holomorphic, eml*-present) at machine precision (MSE $\sim 10^{-32}$); (ii) quantum state evolution, where eml* correctly identifies dissipative systems (7/7 classification); (iii) gravitational lensing ellipticities from KiDS-1000, confirming eml* detection on 5 000 real astrophysical measurements. Negative controls on 175 SPARC galaxy rotation curves, tested with two independent methods (DEAP and PySR), show that eml* produces no signal on purely real data, confirming the method's specificity.

1 Introduction

A fundamental distinction in complex dynamics is between holomorphic maps, which preserve the complex structure, and anti-holomorphic maps, which reverse it through complex conjugation. This distinction governs whether a system conserves information (unitary evolution in quantum mechanics, conformal maps in fluid dynamics) or dissipates it (damped oscillations, measurement collapse).

Despite its importance, no automatic method existed to determine from data alone whether a governing equation is holomorphic or anti-holomorphic. Standard symbolic regression tools [Cranmer, 2023] search over real-valued operators and cannot distinguish z^2 from $\overline{z}^2$ since these coincide on the real line.

Odrzywołek [Odrzywołek, 2026] showed that a single binary operator $\mathrm{eml}(x, y) = \exp(x) - \ln(y)$ generates all elementary functions through uniform binary trees. We extend this framework with $\mathrm{eml}^\star$, the anti-holomorphic companion of eml, and show that it provides the missing piece: an intrinsic marker of anti-holomorphic structure that appears automatically in symbolic regression when the underlying dynamics requires complex conjugation.

2 The $\mathrm{eml}^\star$ operator

Definition 1 (EML operator)**.** *Following Odrzywołek [2026], the EML operator is:*

$$\mathrm{eml}(x, y) = \exp(x) - \ln(y), \quad y \neq 0. \tag{1}$$

Definition 2 ($\mathrm{eml}^\star$ operator)**.** *The anti-holomorphic extension of EML is:*

$$\mathrm{eml}^\star(x, y) = \exp(\overline{x}) - \ln(\overline{y}), \quad y \neq 0. \tag{2}$$

Note that $\mathrm{eml}^\star$ coincides with eml on $\mathbb{R}$, since $\overline{x} = x$ for $x \in \mathbb{R}$. This is not a defect but a feature: $\mathrm{eml}^\star$ can only be distinguished from eml on data with genuine complex structure.

Theorem 1 (Conjugation via $\mathrm{eml}^\star$)**.** *For all $z \in \mathbb{C}$ with $|\mathrm{Im}(z)| < \pi$:*

$$\overline{z} = 1 - \mathrm{eml}^\star(0,\ \mathrm{eml}(z, 1))\,. \tag{3}$$

Proof. We compute stepwise:

$$\begin{align} \mathrm{eml}(z, 1) &= \exp(z) - \ln(1) = \exp(z), \tag{4} \\ \mathrm{eml}^\star(0, \exp(z)) &= \exp(\overline{0}) - \ln(\overline{\exp(z)}) \tag{5} \\ &= 1 - \ln(\exp(\overline{z})) \tag{6} \\ &= 1 - \overline{z}, \tag{7} \end{align}$$

where the last step uses $\ln(\exp(w)) = w$ on the principal branch ($|\mathrm{Im}(w)| < \pi$, satisfied since $\mathrm{Im}(\overline{z}) = -\mathrm{Im}(z)$). Therefore $1 - \mathrm{eml}^\star(0, \mathrm{eml}(z, 1)) = 1 - (1 - \overline{z}) = \overline{z}$. $\square$

Corollary 1 (Modulus squared). $|z|^2 = z \cdot \overline{z} = z \cdot [1 - \mathrm{eml}^\star(0, \mathrm{eml}(z, 1))]$.

Corollary 2 (Density). *The algebra $\mathcal{A}$ generated by $\{\mathrm{eml}, \mathrm{eml}^\star, 1\}$ is dense in $C(K, \mathbb{C})$ for any compact $K \subset \mathbb{C}$ contained in a horizontal strip of height less than 2π, by the Stone–Weierstrass theorem for complex algebras. The algebra separates points (since $\mathrm{eml}(z, 1) = \exp(z)$ is injective on such strips), contains the constants, and is self-adjoint (closed under conjugation, since $\mathrm{eml}^\star$ provides $\overline{z}$ via Theorem 1).*

3 Method

3.1 Symbolic regression with complex operators

We use PySR [Cranmer, 2023] (v1.5.10) with Julia backend, configured for complex arithmetic:

- **Binary operators:** $+, -, \times, \div, \mathrm{eml}, \mathrm{eml}^\star$
- **Unary operators:** $\cos, \sin, \exp, \log, \mathrm{my\_conj}(z) = \overline{z}$, $\mathrm{my\_real}(z) = \mathrm{Re}(z)$, $\mathrm{my\_imag}(z) = \mathrm{Im}(z)$, $\mathrm{my\_abs2}(z) = z\overline{z}$
- **Precision:** `Complex{Float64}`
- **Reproducibility:** `deterministic=True, parallelism="serial", random_state=42`

3.2 Detection criterion

A system is classified as *anti-holomorphic* if the best equation found by PySR contains any of: `emlstar`, `my_conj`, `my_abs2`, or `my_imag`. These all involve complex conjugation.

3.3 Why $\mathrm{eml}^\star$ is necessary

Standard symbolic regression over $\{+, -, \times, \div, \exp, \sin, \cos, \log\}$ cannot distinguish $f(z) = z^2$ from $g(z) = \overline{z}^2$, because all standard operators are holomorphic. Adding $\mathrm{eml}^\star$ to the operator set is the minimal extension that makes anti-holomorphic functions expressible. By Corollary 2, $\{\mathrm{eml}, \mathrm{eml}^\star\}$ suffices to approximate any continuous complex function.

4 Results

4.1 Fractal iteration maps

We generate 200 random points $z \in [-1, 1]^2 \subset \mathbb{C}$ (seed 42) and compute one iteration of three fractal maps with $c = -0.7 + 0.27015i$:

Table 1: Fractal maps: PySR discovers the exact iteration rule and correctly identifies the holomorphic/anti-holomorphic nature.

Fractal	True rule	PySR equation	eml*?	MSE
Mandelbrot	$z^2 + c$	$(x_0 \cdot x_0) - (0.7 - 0.27i)$	No	1.86×10^{-32}
Tricorn	$\overline{z}^2 + c$	my_conj$((x_0 \cdot x_0) - (0.7 + 0.27i))$	**Yes**	2.02×10^{-32}
Burning Ship	$(\lvert\mathrm{Re}\rvert + i\lvert\mathrm{Im}\rvert)^2 + c$	approximation	**Yes**	1.39×10^{-2}

The Mandelbrot and Tricorn results are at machine precision. PySR recovers the exact algebraic form and uses my_conj (equivalent to eml* via Theorem 1) for the Tricorn but not for the Mandelbrot. This is the cleanest possible test: data is natively complex, no encoding artefact is possible, and the only difference between the two maps is one conjugation.

4.2 Quantum state evolution

We generate time series $\psi(t) \to \psi(t+dt)$ for three quantum systems ($\omega = 1.0$, $dt = 0.1$, 300 time steps):

Table 2: Quantum evolution: PySR finds the exact evolution operator and identifies dissipative systems via eml*.

System	$\lvert U\rvert$	eml*?	MSE	Physics
Larmor precession	1.0000	No	5.24×10^{-32}	Unitary
Damped oscillation	0.9900	**Yes**	1.69×10^{-32}	Dissipative
Rabi oscillation	1.0000	No	2.79×10^{-31}	Unitary

All seven systems are classified correctly. eml* appears when the physics requires non-trivial conjugation: dissipation ($|U| < 1$) or measurement ($|z|^2$ with varying z). For the EM plane wave, $|E + iB|^2 = 1$ is constant (the wave is on the unit circle), so PySR discovers the constant directly without conjugation. It does not appear for unitary or conservative systems.

Table 3: Multi-domain physics: additional natively complex systems.

System	Observable	eml*?	MSE
Impedance (RLC circuit)	$\lvert Z(\omega)\rvert^2$	**Yes**	2.48×10^{-31}
Bell state (Born rule)	$\lvert\psi\rvert^2$	**Yes**	1.74×10^{-33}
ECG analytic signal	$\lvert z(t)\rvert^2$	**Yes**	5.58×10^{-34}
EM plane wave	$\lvert E + iB\rvert^2 = 1$ (const.)	No	1.97×10^{-33}

4.3 Astrophysical validation: KiDS-1000

As a first application to real astrophysical data, we use 5 000 galaxy ellipticities from the KiDS-DR4 DEIMOS catalog [Giblin et al., 2021]: each galaxy has a measured shape $\gamma = e_1 + i\, e_2 \in \mathbb{C}$. We test whether PySR recovers $|\gamma|^2 = \gamma \cdot \overline{\gamma}$:

$$\text{PySR finds: my\_abs2}(x_0), \quad \text{MSE} = 2.54 \times 10^{-34}, \quad \text{eml}^\star = \textbf{Yes}. \tag{8}$$

This confirms that eml* operates correctly on real astrophysical measurements with native complex structure. While the test is a sanity check ($|\gamma|^2$ is known to require conjugation), it is the first application of eml* to observational data.

4.4 Negative controls: galaxy rotation curves

To verify specificity, we test eml* on data where no anti-holomorphic signal should be present: the SPARC galaxy rotation curve database [Lelli et al., 2016].

Table 4: Negative controls on galaxy rotation curves (real-valued data). eml* produces no meaningful signal, as expected.

Method	Galaxies	eml* rate	Interpretation
DEAP (Option A: $z = r + 0i$)	124	24%	Combinatorial noise
PySR (Option B: $z = r + iv_{\text{bar}}$)	171	88%	Encoding artefact

DEAP (Option A): input is purely real ($z = r + 0i$). Since $\overline{z} = z$ for $z \in \mathbb{R}$, eml* = eml and the 24% rate reflects random operator selection by the genetic algorithm ($\sim 1/k$ for k binary operators). Examination of all 30 "eml*-positive" equations confirms that conjugation acts on real values or constants, making it an identity operation.

PySR (Option B): input is artificially complexified ($z = r_{\mathrm{norm}} + i\, v_{\mathrm{bar,norm}}$). The inflated 88% rate arises because PySR uses my_real and my_imag to extract the real target from the complex input—counted as eml* but reflecting encoding, not physics.

These controls demonstrate that eml* detection is specific to genuinely complex data and produces no false positives on real-valued inputs.

5 Discussion

5.1 What eml* detects

Across all validated tests, eml* appears when the dynamics involves complex conjugation and does not appear when it does not:

- Anti-holomorphic maps ($\overline{z}^2 + c$)
- Dissipative evolution ($|U| < 1$, requiring $|z|^2 = z\overline{z}$)
- Born rule measurements ($P = |\psi|^2$)

It does not appear for holomorphic, unitary, or energy-conserving systems.

5.2 Limitations

Branch cut restriction. Theorem 1 requires $|\mathrm{Im}(z)| < \pi$ for the principal branch of the logarithm. All data in this work satisfies this condition; extensions to $|\mathrm{Im}(z)| \geq \pi$ would require branch-tracking.

Real-valued data. eml* is structurally identical to eml on $\mathbb{R}$. The method is only informative on data with genuine complex structure. This is a mathematical fact, not a limitation of the implementation.

Neural world model. We attempted to build a JEPA-style neural network (complex-valued, with separate holomorphic and anti-holomorphic pathways). In a supervised setting, the model achieves 88% classification accuracy, but without supervision, the learned gate does not differentiate the two pathways. The root cause is that $|\overline{z}| = |z|$: a gate that receives only magnitudes cannot distinguish holomorphic from anti-holomorphic dynamics. This remains an open problem.

5.3 Coordinate representations

We tested eml* on the Schwarzschild metric in two formulations: the standard Hilbert form (which crosses the horizon $r = 2M$ and produces complex values) yields eml* = True, while the original Schwarzschild form (which avoids the interior) yields eml* = False. This is a result about coordinate representations, not about physical reality: the coordinate singularity at $r = 2M$ has been understood since Kruskal (1960).

6 Conclusion

We have shown that the eml* operator, combined with symbolic regression, provides an automatic, interpretable criterion for detecting anti-holomorphic structure in complex dynamical systems. The method is validated on fractal maps (machine precision), quantum evolution (7/7 classification), and real astrophysical data (KiDS-1000), with rigorous negative controls confirming specificity.

The eml* framework opens several directions: (i) application to natively complex astrophysical data (gravitational lensing shear fields, interferometric visibilities); (ii) integration as a structural prior in neural world models; (iii) extension to quaternionic or Clifford-algebraic settings.

All code and data are available at `https://github.com/antparis/oxieml-star`.

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